5G RAN

Extended Cell Range Feature Parameter Description

Issue Draft A

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters
- Editorial changes

  Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-11-07).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for dual-period 8:2 slot configuration by Extended Cell Range (TDD) and Extended Cell Range of 90 km (TDD). For details, see: • **3.2 Random Access Preamble Format** • **3.3 Slot Configuration and Slot Structure** • **4.3.2.1 Prerequisite Functions** • **4.3.2.2 Mutually Exclusive Functions** • **4.3.2.3 Function Impacts** • **4.3.3 Hardware** • **4.3.5 Cells** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands** • **5.3.2.1 Prerequisite Functions** • **5.3.2.2 Mutually Exclusive Functions** • **5.3.2.3 Function Impacts** • **5.4.1.1 Data Preparation** • **5.4.1.2 Using MML Commands**	None	TDD	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Added descriptions of the theoretical maximum cell radius corresponding to the number of GP symbols for slot structures SS55 and SS85. For details, see **3.3 Slot Configuration and Slot Structure**.

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

Feature ID	Feature Name	Chapter/Section
FOFD-051201	Extended Cell Range	**4 Extended Cell Range (TDD)** **5 Extended Cell Range of 90 km (TDD)**

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Extended Cell Range	None	**4 Extended Cell Range (TDD)** **5 Extended Cell Range of 90 km (TDD)**
Directional outmigration of cell-center UEs from an extended-range cell	Supported only in SA networking	**6 Directional Outmigration of Cell-Center UEs from an Extended-Range Cell**

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Extended Cell Range	Supported only in low frequency bands	**4 Extended Cell Range (TDD)** **5 Extended Cell Range of 90 km (TDD)**

Function Name	Difference	Chapter/Section
Directional outmigration of cell-center UEs from an extended-range cell	Supported only in low frequency bands	**6 Directional Outmigration of Cell-Center UEs from an Extended-Range Cell**

3 Overview

Today's mobile communication networks cover almost all inland areas. However, in areas such as open seas, deserts, and open plains, site selection presents major challenges due to sparse populations and hostile environmental conditions. For example, finding suitable site locations at sea can be difficult, and deploying standard sites in deserts or in open plains (where traffic is light) can be prohibitively expensive. To reduce costs and ensure network coverage in these areas, solutions for extended-range scenarios are introduced.

3.1 Definition

Extended-range scenarios refer to scenarios with a larger coverage area than common scenarios. Solutions for extended-range scenarios provide Extended Cell Range and directional outmigration of cell-center UEs from an extended-range cell.

- Extended Cell Range aims to expand the cell coverage and includes Extended Cell Range (TDD) and Extended Cell Range of 90 km (TDD).
  - When Extended Cell Range (TDD) is enabled, the theoretical cell radius is collectively determined by two factors: (1) access capability corresponding to the random access preamble format (see **3.2 Random Access Preamble Format**); (2) slot configuration and slot structure (see **3.3 Slot Configuration and Slot Structure**). The smaller value between the cell radii corresponding to the two factors is regarded as the theoretical maximum cell radius, which can reach up to 60 km.
  - When Extended Cell Range of 90 km (TDD) is enabled, the theoretical cell radius is collectively determined by two factors: (1) access capability corresponding to the random access preamble format; (2) slot configuration and slot structure. The smaller value between the cell radii corresponding to the two factors is regarded as the theoretical maximum cell radius, which can reach up to 90 km.
- Directional outmigration of cell-center UEs from an extended-range cell aims to improve cell-edge user experience, especially in sea areas where sites are not high-altitude ones. This function defines cells with a radius greater than 3 km as extended-range cells and does not depend on Extended Cell Range.

Table 3-1 Functions in this document as well as their implementation and application scenarios

Function	Implementation	Application Scenario	Chapter/Section
Extended Cell Range (TDD)	This function uses preamble format 0 and optimizes PRACH demodulation performance to improve the access capability corresponding to the random access preamble format, enabling the maximum cell radius to reach 60 km. (PRACH is short for physical random access channel.)	The **NRDUCell**.***CellRadius*** parameter is set to a value (in the unit of m) greater than **14500** and less than or equal to **60000**.	**4 Extended Cell Range (TDD)**
Extended Cell Range of 90 km (TDD)	This function uses preamble format 3 to further improve the access capability corresponding to the random access preamble format since the preamble sequence is repeated multiple times in this format. In addition, a self-contained slot includes 17 or 18 guard period (GP) symbols. As a result, the maximum cell radius can reach 90 km.	The **NRDUCell**.***CellRadius*** parameter is set to a value (in the unit of m) greater than **60000** and less than or equal to **90000**.	**5 Extended Cell Range of 90 km (TDD)**
Directional outmigration of cell-center UEs from an extended-range cell	This function identifies cell-center UEs in an extended-range cell and migrates these UEs directionally to surrounding cells. This allows the extended-range cell to preferentially serve its cell-edge UEs.	The **NRDUCell**.***CellRadius*** parameter is set to a value (in the unit of m) greater than **3000**.	**6 Directional Outmigration of Cell-Center UEs from an Extended-Range Cell**

NOTE

In this document, the maximum cell radius refers to the theoretical maximum cell radius with Extended Cell Range enabled. The actual cell radius is generally less than the theoretical maximum value.

3.2 Random Access Preamble Format

Random access preamble formats are defined in section 6.3.3 of 3GPP TS 38.211 (Release 15) to satisfy the various coverage requirements of wireless networks. A preamble format determines the time domain resources available for a PRACH. In a random access procedure, the gNodeB calculates the maximum supported cell radius based on the preamble format transmitted from a UE.

To enable Extended Cell Range for a cell, the cell must use an appropriate random access preamble format. The preamble format is specified by the **NRDUCellPrach**.***PrachConfigurationIndex*** parameter.

Preamble format 0 is used when Extended Cell Range (TDD) is enabled. In this case, the parameter must be set to **9**, **12**, **14**, **17**, **19**, **65535**, or a value ranging from **2** to **7**.

Preamble format 3 is used when Extended Cell Range of 90 km (TDD) is enabled. In this case, the parameter must be set to **49**, **52**, **54**, **57**, **59**, **65535**, or a value ranging from **43** to **47**.

3.3 Slot Configuration and Slot Structure

In NR TDD, the maximum cell radius can be increased by changing the slot configuration and slot structure. The larger the number of GP symbols in self-contained slots, the larger the maximum cell radius. **Table 3-2** lists the mapping between the number of GP symbols and cell coverage capability for different slot configurations and slot structures, without considering the preamble format.

As such, in NR TDD, to enable Extended Cell Range for a cell, the cell must be configured with an appropriate slot configuration and slot structure.

- Extended Cell Range (TDD) extends cell coverage by increasing the number of GP symbols and using one of the following combinations:
  - Single-period 8:2 slot configuration (with the **NRDUCell**.***SlotAssignment*** parameter set to **8_2_DDDDDDDSUU**) and slot structure SS56 or SS518 (with the **NRDUCell**.***SlotStructure*** parameter set to **SS56** or **SS518**)
  - Dual-period 8:2 slot configuration (with the **NRDUCell**.***SlotAssignment*** parameter set to **8_2_DDDSUUDDDD**) and slot structure SS86 or SS818 (with the **NRDUCell**.***SlotStructure*** parameter set to **SS86** or **SS818**)
- Extended Cell Range of 90 km (TDD) extends cell coverage by increasing the number of GP symbols and using one of the following combinations:
  - Single-period 8:2 slot configuration (with the **NRDUCell**.***SlotAssignment*** parameter set to **8_2_DDDDDDDSUU**) and slot structure SS518 (with the **NRDUCell**.***SlotStructure*** parameter set to **SS518**)
  - Dual-period 8:2 slot configuration (with the **NRDUCell**.***SlotAssignment*** parameter set to **8_2_DDDSUUDDDD**) and slot structure SS818 (with the **NRDUCell**.***SlotStructure*** parameter set to **SS818**)

Table 3-2 Mapping between the number of GP symbols and theoretical maximum cell radius for different slot configurations and slot structures in low frequency bands

Slot Configuration and Slot Structure	Number of GP Symbols	Theoretical Maximum Cell Radius (km)
Single-period 8:2 slot configuration with slot structure SS51 or dual-period 8:2 slot configuration with slot structure SS81	1	1.4
Single-period 8:2 slot configuration with slot structure SS52 or dual-period 8:2 slot configuration with slot structure SS82	2	6.8
Single-period 8:2 slot configuration with slot structure SS53 or dual-period 8:2 slot configuration with slot structure SS83	3	12.2
Single-period 8:2 slot configuration with slot structure SS54 or dual-period 8:2 slot configuration with slot structure SS84	4	17.5
Single-period 8:2 slot configuration with slot structure SS55 or dual-period 8:2 slot configuration with slot structure SS85	5	22.9
Single-period 8:2 slot configuration with slot structure SS56 or dual-period 8:2 slot configuration with slot structure SS86	6	28.2
Single-period 8:2 slot configuration with slot structure SS518 or dual-period 8:2 slot configuration with slot structure SS818	18	92.4

NOTE

1. For details about the slot configuration and slot structure, see *Standards Compliance*.
2. In scenarios where both LTE and NR are deployed, the slot structures of LTE and NR must ensure that the duration of GP symbols is consistent between the two RATs. For example, if the number of GP symbols on the NR side is set to 18, the number of GP symbols on the LTE side must be set to 9.

4 Extended Cell Range (TDD)

4.1 Principles

For details, see **3 Overview**.

4.2 Network Analysis

4.2.1 Benefits

The coverage of the base station is expanded. Theoretically, the maximum cell radius can reach 60 km.

NOTE

Whether the maximum cell radius can reach 60 km depends on the factors responsible for path loss difference, such as the base station's transmit power and antenna altitude as well as the UE's antenna gains.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

More cell-edge UEs may access cells since the coverage of the base station is expanded. An increased number of such UEs will negatively impact the indicators related to access, handovers, and service drops, as well as specific indicators such as the **User Uplink Average Throughput (DU)**, **User Downlink Average Throughput (DU)**, **Cell Uplink Average Throughput (DU)**, and **Cell Downlink Average Throughput (DU)**.

4.3 Requirements

4.3.1 Licenses

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td>FOFD-051201</td><td>Extended Cell Range</td><td>NR0S00EXCR00</td><td>Per Cell</td></tr>
<tr><td colspan="4">For the license control item NR0S00EXCR00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
– When the license sales number pre-check function is enabled, the NRDUCell.CellRadius parameter cannot be modified due to license insufficiency, preventing cell activation failures.
– When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.
The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	Extended Cell Range (TDD) requires that the **NRDUCell**.***SlotAssignment*** parameter be set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
Slot structure	**NRDUCell**.***SlotStructure***	*Standards Compliance*	Extended Cell Range (TDD) supports only SS56, SS518, SS86, and SS818.
3D coverage pattern	**NRDUCellTrpBeam**.***CoverageScenario***	*Beam Management*	When the **NRDUCell**.***SlotStructure*** parameter is set to **SS518** or **SS818**, the **NRDUCellTrpBeam**.***CoverageScenario*** parameter must be set to **DEFAULT**, **EXPAND_SCENARIO_7**, **EXPAND_SCENARIO_8**, **SCENARIO_27**, **SCENARIO_28**, **SCENARIO_31**, **SCENARIO_33**, **SCENARIO_34**, **SCENARIO_35**, **SCENARIO_36**, **CUSTOMIZED_BEAM_SCENARIO**, or a value ranging from **SCENARIO_1** to **SCENARIO_16**.

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	None
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	None

Function Name	Function Switch	Reference	Description
Remote interference adaptive avoidance	**RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	None
Enhanced remote interference adaptive avoidance	**RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	None
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	None
Distributed massive MIMO	**DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	None
Low power high accuracy positioning	**LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	None
Inactive RA-SDT	**REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*RedCap*	When Extended Cell Range (TDD) is enabled, the **REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	Co-MM cannot be enabled when the **NRDUCell**.***SlotStructure*** parameter is set to **SS518** or **SS818**.

Function Name	Function Switch	Reference	Description
Inactive RA-SDT	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*UE Power Saving*	If the **NRDUCell**.***CellRadius*** parameter is set to a value greater than **14500**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
Uplink symbol power saving	**UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	None

4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Remote Interference Avoidance for SRS Sw	**SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	When the switch for SRS remote interference avoidance is turned on, the PRACH guard time is shortened, which may decrease the random access success rate of UEs on which Extended Cell Range takes effect.
Remote interference resistance for random access	**RA_REMOTE_INTRF_RESIST_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	Remote interference resistance for random access restricts the access of cell-edge UEs. As a result, cell-edge UEs may fail to access the network when Extended Cell Range is enabled.

Function Name	Function Switch	Reference	Description
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When proactive avoidance of remote interference is enabled, the PRACH guard time is shortened, which may decrease the random access success rate of UEs on which Extended Cell Range takes effect.
Clock out-of-synchronization detection	**CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter	*Synchronization*	If the switch for clock out-of-synchronization detection and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled, the positions for sending and detecting synchronization sequences will be adjusted. As a result, synchronization sequence detection will fail between the base station serving these cells and a neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled.

Function Name	Function Switch	Reference	Description
Air-interface-based network-wide synchronization deviation detection	**GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter	*Synchronization*	If the switch for air-interface-based network-wide synchronization deviation detection and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled, the positions for sending and detecting synchronization sequences will be adjusted. As a result, sequence detection will fail between the base station serving these cells and a neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled.

Function Name	Function Switch	Reference	Description
Ultra-long holdover for sites with abnormal clock sources	**CLK_OTASYNC_PROC_SW** option of the **TASM**.***MSCLKDIAGPROCESSSW*** parameter	*Synchronization*	If the switch for ultra-long holdover for sites with abnormal clock sources and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled, the positions for sending and detecting synchronization sequences will be adjusted. As a result, sequence detection will fail between the base station serving these cells and a neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	If Fusion Cell is required to be enabled in a cell with Extended Cell Range enabled and the **NRDUCell**.***SlotStructure*** parameter set to **SS518** or **SS818**, then the **SS518_4GP_TIME_INTRF_AVOID_SW** or **SS818_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter corresponding to the cell must be selected.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards as well as NR TDD-capable UBBPg and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

When the **NRDUCell**.***SlotAssignment*** parameter is set to **8_2_DDDSUUDDDD**, all NR TDD-capable massive MIMO RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

When the **NRDUCell**.***SlotAssignment*** parameter is set to a value other than **8_2_DDDSUUDDDD**, all NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

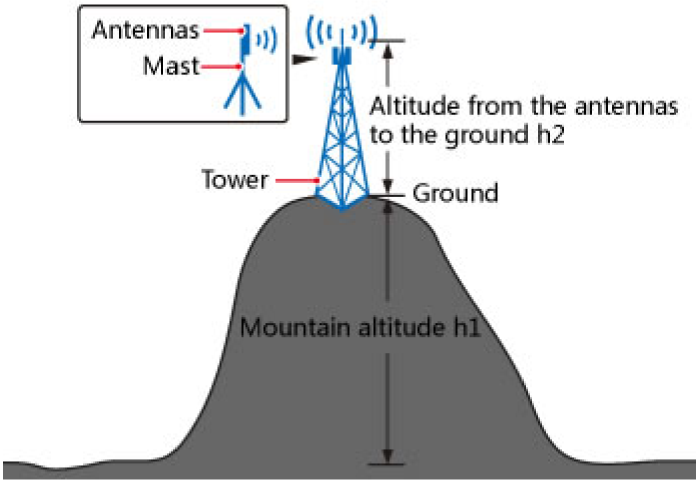
4.3.4 Networking

It is recommended that the following networking planning policies be used to ensure better service rates at the cell edge when Extended Cell Range is enabled:

- Antenna altitude

A valid antenna altitude consists of two parts: mountain altitude (*h1*) and the altitude from the antennas to the ground (*h2*), as shown in **Figure 4-1**. Due to limitations on the altitude of a tower or mast where the antennas are mounted, the maximum altitude from the antennas to the ground is 70 m.

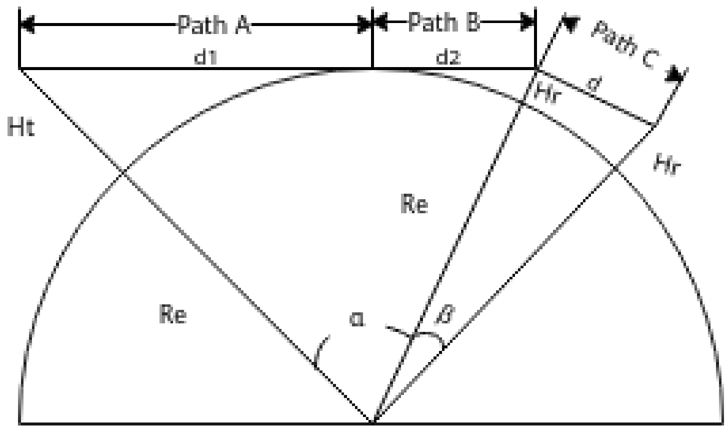
Figure 4-1 Antenna altitude



The valid antenna altitude determines the maximum cell radius on a live network. Therefore, ensure that the target cell coverage is within LOS when planning the antenna altitude. NLOS areas may experience signal attenuation and poor cell coverage due to the Earth's curvature.

For ocean coverage, Huawei provides a segmentation model to describe radio propagation on the sea, as shown in **Figure 4-2**.

Figure 4-2 Segmentation model for radio propagation on the sea



As shown in the preceding figure:

- *Ht* denotes the site altitude above sea level.
- *Hr* denotes the UE altitude above sea level.
- *d1* denotes the LOS path A on the sea.
- *d2* denotes the LOS path B on the sea.
- *Re* denotes the Earth's radius.
- *d* denotes path C, which is a shadow area with poor signal quality due to the Earth's curvature. Therefore, this path can be ignored during network planning.

According to the model for radio propagation on the sea, **Table 4-1** lists the LOS distances corresponding to different site and UE altitudes.

Table 4-1 LOS distances corresponding to different site and UE altitudes

Ht (m)	d1 + d2 (km) when Hr is 0 m	d1 + d2 (km) when Hr is 3 m	d1 + d2 (km) when Hr is 5 m	d1 + d2 (km) when Hr is 10 m	d1 + d2 (km) when Hr is 30 m
20	18.4	25.6	27.6	31.5	41
30	22.6	29.7	31.8	35.6	45.1
50	29.1	36.3	38.3	42.2	51.7
80	36.9	39.0	46.1	49.9	59.4
100	41.2	44	50.4	54.2	63.8
200	58.3	57.6	67.5	71.3	80.8
300	71.4	72.3	80.6	84.4	93.9
400	82.4	89.5	91.6	95.4	105

When deploying Extended Cell Range at sea, calculate the antenna altitude based on the LOS distances (d1 + d2) listed in the preceding tables.

- Site acquisition

  As the deployment of Extended Cell Range requires good radio propagation, the following requirements must be met for site acquisition:

  – There are no obvious obstacles in the target area to be covered by the base station, which is within LOS.

  – Site altitude is as high as possible based on coverage requirements.

  – The GSM, UMTS, or LTE sites enabled with Extended Cell Range are preferentially selected for deploying NR Extended Cell Range.

  – For other requirements, see related requirements on macro networks.

- Antenna model selection

  – Antenna gain: In 8T8R scenarios, high-gain common antennas are recommended.

  – Antenna polarization: Vertically polarized antennas are recommended since they deliver better performance than other types of polarized antennas in open areas.

  – Antenna surface area: Antennas with smaller surface areas are recommended in order to reduce wind resistance on the tower or mast where they are mounted.

  – Antenna waveform: When Extended Cell Range is deployed at sea, the antennas are mounted as high as possible in order to resist the effect of the Earth's curvature. However, this may also create coverage holes at the near end. In this scenario, antennas with suitable vertical beamwidths must be selected to avoid such coverage holes.

– Tilt adjustment: As Extended Cell Range has few requirements on tilt adjustment, non-remote-electrical-tilt (non-RET) antennas are recommended. At extremely high altitude sites, electrical and mechanical downtilts can be adjusted to improve local cell coverage, while continuing to ensure remote cell coverage.

4.3.5 Cells

- When the **NRDUCell**.***SlotAssignment*** parameter is set to **8_2_DDDSUUDDDD** for a cell, Extended Cell Range (TDD) requires the **NRDUCellTrp**.***TxRxMode*** parameter to be set to **64T64R** or **32T32R** for the cell.
- When the **NRDUCell**.***SlotAssignment*** parameter is set to a value other than **8_2_DDDSUUDDDD** for a cell, Extended Cell Range (TDD) requires the **NRDUCellTrp**.***TxRxMode*** parameter to be set to **64T64R**, **32T32R**, **8T8R**, **4T4R**, **2T2R**, or **1T1R** for the cell.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Cell Radius	**NRDUCell**.***CellRadius***	Set this parameter based on the network plan. Extended Cell Range (TDD) takes effect when this parameter is set to a value greater than **14500** (unit: m) and less than or equal to **60000** (unit: m).
Slot Assignment	**NRDUCell**.***SlotAssignment***	Set this parameter to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
Slot Structure	**NRDUCell**.***SlotStructure***	Set this parameter to **SS56**, **SS518**, **SS86**, or **SS818**.
PRACH Configuration Index	**NRDUCellPrach**.***PrachConfigurationIndex***	Set this parameter to **9**, **12**, **14**, **17**, **19**, **65535**, or a value ranging from **2** to **7**.

Table 4-3 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Channel Calibration Policy Switch	**ADAPTIVE_ADJ_SW** option of the **gNodeBAlgo**.***ChnCalibPolSw*** parameter	It is recommended that this option be selected when Extended Cell Range is enabled.

Parameter Name	Parameter ID	Setting Notes
Time Interference Avoidance Switch	**gNodeBAlgo**.***TimeIntrfAvoidSw***	In scenarios where cells with Extended Cell Range enabled and common cells provide contiguous coverage: ● Assume that the **NRDUCell**.***SlotStructure*** parameter is set to **SS54** for common cells. – When the **NRDUCell**.***SlotStructure*** parameter is set to **SS56** for cells with Extended Cell Range enabled, the **SS56_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter needs to be selected to enable time-domain interference avoidance. – When the **NRDUCell**.***SlotStructure*** parameter is set to **SS518** for cells with Extended Cell Range enabled, the **SS518_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter needs to be selected to enable time-domain interference avoidance. ● Assume that the **NRDUCell**.***SlotStructure*** parameter is set

Parameter Name	Parameter ID	Setting Notes
		to **SS84** for common cells. – When the **NRDUCell**.***SlotStructure*** parameter is set to **SS86** for cells with Extended Cell Range enabled, the **SS86_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter needs to be selected to enable time-domain interference avoidance. – When the **NRDUCell**.***SlotStructure*** parameter is set to **SS818** for cells with Extended Cell Range enabled, the **SS818_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter needs to be selected to enable time-domain interference avoidance. After an option of the parameter is selected in scenarios where both LTE and NR are deployed, the number of GP symbols on the LTE side can only be set to 2.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

Modifying any one of the following parameters will cause the cell to restart: **NRDUCell**.***CellRadius***, **NRDUCell**.***SlotAssignment***, **NRDUCell**.***SlotStructure***, **NRDUCellPrach**.***PrachConfigurationIndex***, and **gNodeBAlgo**.***TimeIntrfAvoidSw***.

Activation Command Examples

```
//Restoring the PRACH configuration index to its default value if it is not set to 9, 12, 14, 17, 19, 65535, or a value ranging from 2 to 7
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=65535;
//Setting the slot configuration and slot structure (for example, when the frequency band is n41 and the number of GP symbols is 18)
MOD NRDUCELL: FrequencyBand=N41, DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ, SlotAssignment=8_2_DDDDDDDSUU, SlotStructure=SS518, NrDuCellId=0, CellId=0;
//Setting the maximum cell radius according to the network plan
MOD NRDUCELL: NrDuCellId=0, CellRadius=60000;
//Configuring PRACH preamble format 0
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=17;
```

Optimization Command Examples

```
//Turning on the switch for adaptive adjustment
MOD GNODEBALGO: ChnCalibPolSw=ADAPTIVE_ADJ_SW-1;
//Turning on the switch for time-domain interference avoidance (for example, when the slot structures of common cells and cells with Extended Cell Range enabled are SS54 and SS518, respectively)
MOD GNODEBALGO: TimeIntrfAvoidSw=SS518_4GP_TIME_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Restoring the PRACH configuration index to its default value
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=65535;
//Disabling Extended Cell Range
MOD NRDUCELL: NrDuCellId=0, CellRadius=10000;
//Restoring the slot configuration and slot structure (for example, when the frequency band is n41 and the number of GP symbols is 4)
MOD NRDUCELL: FrequencyBand=N41, DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ, SlotAssignment=8_2_DDDDDDDSUU, SlotStructure=SS54, NrDuCellId=0, CellId=0;
```

4.4.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

To verify whether Extended Cell Range (TDD) has taken effect, perform the following operations:

1. Run the **LST NRDUCELL** command to check whether the **NRDUCell**.***CellRadius*** parameter value is greater than **14500**.
2. Run the **DSP NRDUCELL** command to check the cell status and verify that the cell has been activated.

4.4.4 Network Monitoring

None

5 Extended Cell Range of 90 km (TDD)

5.1 Principles

For details, see **3 Overview**.

5.2 Network Analysis

5.2.1 Benefits

When Extended Cell Range of 90 km (TDD) is in effect, the coverage of the base station is expanded. Theoretically, the maximum cell radius can reach 90 km.

NOTE

Whether the maximum cell radius can reach 90 km depends on the factors responsible for path loss difference, such as the base station's transmit power and antenna altitude as well as the UE's antenna gains.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

Extended Cell Range of 90 km (TDD) has the following impacts when it is enabled in a cell with Extended Cell Range disabled:

- More cell-edge UEs may access cells since the coverage of the base station is expanded. An increased number of such UEs will negatively impact the indicators such as the access success rate and service drop rate, as well as specific indicators such as the **User Uplink Average Throughput (DU)**, **User Downlink Average Throughput (DU)**, **Cell Uplink Average Throughput (DU)**, and **Cell Downlink Average Throughput (DU)**.
- This function requires that the number of GP symbols be 17 or 18. As a result, the available number of symbols and RB resources for downlink scheduling decrease, the downlink cell peak rate and average downlink cell throughput slightly decrease, and the delay slightly increases.

- Since the number of GP symbols increases, the number of SSB beams decreases and the broadcast channel overhead decreases.
- Since the number of GP symbols increases, UE-level CSI beams do not take effect. For large-packet UEs, the average downlink UE throughput decreases significantly.
- Since the number of GP symbols increases, the positions of slots for RAR delivery change. The difference in bit error rates (BERs) between slots causes the contention-based and non-contention-based access success rates to fluctuate.
- Since the PRACH preamble format is changed from format 0 to format 3, frequency-domain resource consumption increases. As a result, the uplink cell peak rate and average uplink cell throughput decrease, and the uplink PRB usage increases.
- UEs that do not support long PRACH preamble format 3 (if any) experience access failures or handover failures due to UE compatibility issues.
- If cell-edge UEs support format 3 but the maximum timing advance (TA) value is less than 2308 (corresponding to a distance of 90 km), they may experience access failures or handover failures.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051201	Extended Cell Range	NR0S00EXCR00	Per Cell

For the license control item NR0S00EXCR00:

- If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
- If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
  - When the license sales number pre-check function is enabled, the **NRDUCell**.***CellRadius*** parameter cannot be modified due to license insufficiency, preventing cell activation failures.
  - When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.

The license sales number pre-check function is controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld**.***LicenseChkSw*** parameter.

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	Extended Cell Range of 90 km (TDD) requires that the **NRDUCell**.***SlotAssignment*** parameter be set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
Slot structure	**NRDUCell**.***SlotStructure***	*Standards Compliance*	Extended Cell Range of 90 km (TDD) requires that the **NRDUCell**.***SlotStructure*** parameter be set to **SS518** or **SS818**.
PRACH Configuration Index	**NRDUCellPrach**.***PrachConfigurationIndex***	None	Extended Cell Range of 90 km (TDD) requires that the **NRDUCellPrach**.***PrachConfigurationIndex*** parameter be set to **49**, **52**, **54**, **57**, **59**, **65535**, or a value ranging from **43** to **47**.

Function Name	Function Switch	Reference	Description
3D coverage pattern	**NRDUCellTrpBeam**.***CoverageScenario***	*Beam Management*	When Extended Cell Range of 90 km (TDD) is enabled and the **NRDUCell**.***SlotStructure*** parameter is set to **SS518** or **SS818**, then the **NRDUCellTrpBeam**.***CoverageScenario*** parameter must be set to **DEFAULT**, **EXPAND_SCENARIO_7**, **EXPAND_SCENARIO_8**, **SCENARIO_27**, **SCENARIO_28**, **SCENARIO_31**, **SCENARIO_33**, **SCENARIO_34**, **SCENARIO_35**, **SCENARIO_36**, **CUSTOMIZED_BEAM_SCENARIO**, or a value ranging from **SCENARIO_1** to **SCENARIO_16**.

5.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	None
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	None

Function Name	Function Switch	Reference	Description
Remote interference adaptive avoidance	**RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	None
Enhanced remote interference adaptive avoidance	**RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	None
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	None
PUSCH Share PRACH Freq Resource Sw	**PUSCH_SHR_PRACH_FREQ_RES_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	None	None
Inactive RA-SDT	**REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*RedCap*	When Extended Cell Range of 90 km (TDD) is enabled, the **REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
Distributed massive MIMO	**DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	None

Function Name	Function Switch	Reference	Description
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	Co-MM cannot be enabled when the **NRDUCell**.***SlotStructure*** parameter is set to **SS518** or **SS818**. Therefore, Extended Cell Range of 90 km (TDD) cannot be enabled with Co-MM.
Low power high accuracy positioning	**LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	None
Inactive RA-SDT	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*UE Power Saving*	If the **NRDUCell**.***CellRadius*** parameter is set to a value greater than **14500**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
Uplink symbol power saving	**UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	None

5.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Remote Interference Avoidance for SRS Sw	**SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	When the switch for SRS remote interference avoidance is turned on, the PRACH guard time is shortened, which may decrease the random access success rate and cause access failures for UEs on which Extended Cell Range of 90 km (TDD) takes effect. Therefore, it is not recommended that the switch be turned on when Extended Cell Range of 90 km (TDD) is enabled.
Remote interference resistance for random access	**RA_REMOTE_INTRF_RESIST_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	Remote interference resistance for random access restricts the access of cell-edge UEs. As a result, cell-edge UEs may fail to access the network when Extended Cell Range of 90 km (TDD) is enabled.
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When proactive avoidance of remote interference is enabled, the PRACH guard time is shortened, which may decrease the random access success rate and cause access failures for UEs on which Extended Cell Range of 90 km (TDD) takes effect. Therefore, it is not recommended that the two functions be both enabled.

Function Name	Function Switch	Reference	Description
Clock out-of-synchronization detection	**CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter	*Synchronization*	If the switch for clock out-of-synchronization detection and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled and the slot structure set to SS518, the positions for sending and detecting synchronization sequences will be adjusted. As a result, synchronization sequence detection will fail between the base station serving these cells and a neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled and the slot structure set to SS518.

Function Name	Function Switch	Reference	Description
Air-interface-based network-wide synchronization deviation detection	**GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter	*Synchronization*	If the switch for air-interface-based network-wide synchronization deviation detection and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled and the slot structure set to SS518, the positions for sending and detecting synchronization sequences will be adjusted. As a result, sequence detection will fail between the base station serving these cells and a neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled and the slot structure set to SS518.

Function Name	Function Switch	Reference	Description
Ultra-long holdover for sites with abnormal clock sources	**CLK_OTASYNC_PROC_SW** option of the **TASM**.***MSCLKDIAGPROCESSSW*** parameter	*Synchronization*	If the switch for ultra-long holdover for sites with abnormal clock sources and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled and the slot structure set to SS518, the positions for sending and detecting synchronization sequences will be adjusted. As a result, sequence detection will fail between the base station serving these cells and a neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled and the slot structure set to SS518.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	If Fusion Cell is required to be enabled in a cell with Extended Cell Range of 90 km (TDD) enabled, then the **SS518_4GP_TIME_INTRF_AVOID_SW** or **SS818_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter corresponding to the cell must be selected.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards as well as NR TDD-capable UBBPg and later boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable massive MIMO AAUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Networking

For details, see **4.3.4 Networking**.

5.3.5 Cells

Extended Cell Range of 90 km (TDD) requires low-frequency TDD massive MIMO cells.

5.3.6 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Cell Radius	**NRDUCell**.***CellRadius***	Set this parameter based on the network plan. Extended Cell Range of 90 km (TDD) takes effect when this parameter is set to a value greater than **60000** (unit: m) and less than or equal to **90000** (unit: m).
Slot Assignment	**NRDUCell**.***SlotAssignment***	Set this parameter to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
Slot Structure	**NRDUCell**.***SlotStructure***	Set this parameter to **SS518** or **SS818**.
PRACH Configuration Index	**NRDUCellPrach**.***PrachConfigurationIndex***	Set this parameter to **49**, **52**, **54**, **57**, **59**, **65535**, or a value ranging from **43** to **47**.
Channel Calibration Policy Switch	**ADAPTIVE_ADJ_SW** option of the **gNodeBAlgo**.***ChnCalibPolSw*** parameter	It is recommended that this option be selected when Extended Cell Range of 90 km (TDD) is enabled and strong interference exists due to differences in the number of GP symbols in the slot structures between common cells and cells with Extended Cell Range enabled.

Parameter Name	Parameter ID	Setting Notes
Time Interference Avoidance Switch	**gNodeBAlgo**.*TimeIntrfAvoidSw*	In scenarios where cells with Extended Cell Range enabled and common cells provide contiguous coverage and strong interference exists due to differences in the number of GP symbols in the slot structures between these two types of cells: ● Assume that the **NRDUCell**.*SlotStructure* parameter is set to **SS54** for common cells. The **SS518_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.*TimeIntrfAvoidSw* parameter needs to be selected to enable time-domain interference avoidance. ● Assume that the **NRDUCell**.*SlotStructure* parameter is set to **SS84** for common cells. The **SS818_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo**.*TimeIntrfAvoidSw* parameter needs to be selected to enable time-domain interference avoidance. After an option of the parameter is selected in scenarios where both LTE and NR are deployed, the number of GP symbols on the LTE side can only be set to 2.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

Modifying any one of the following parameters will cause the cell to restart: **NRDUCell**.***CellRadius***, **NRDUCell**.***SlotAssignment***, **NRDUCell**.***SlotStructure***, **NRDUCellPrach**.***PrachConfigurationIndex***, and **gNodeBAlgo**.***TimeIntrfAvoidSw***.

Activation Command Examples

```
//Restoring the PRACH configuration index to its default value if it is not set to 49, 52, 54, 57, 59, 65535, or a value ranging from 43 to 47
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=65535;
//Configuring the slot configuration and slot structure (for example, when the slot configuration is single-period 8:2 and the number of GP symbols is 18)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, CellId=0, FrequencyBand=N41, SubcarrierSpacing=30KHZ, SlotAssignment=8_2_DDDDDDDSUU, SlotStructure=SS518;
//Setting the cell radius as planned to enable Extended Cell Range of 90 km (TDD)
MOD NRDUCELL: NrDuCellId=0, CellRadius=70000;
//Configuring PRACH preamble format 3
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=57;
//(Optional) Turning on the switch for adaptive adjustment when strong interference exists due to differences in the number of GP symbols in the slot structures between common cells and cells with Extended Cell Range enabled
MOD GNODEBALGO: ChnCalibPolSw=ADAPTIVE_ADJ_SW-1;
//(Optional) Turning on the related switch for time-domain interference avoidance when strong interference exists due to differences in the number of GP symbols in the slot structures between common cells and cells with Extended Cell Range enabled (for example, when the slot structures of common cells and cells with Extended Cell Range enabled are SS54 and SS518, respectively)
MOD GNODEBALGO: TimeIntrfAvoidSw=SS518_4GP_TIME_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Restoring the PRACH format configuration, that is, configuring PRACH preamble format 0
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=65535;
//Disabling Extended Cell Range of 90 km (TDD)
MOD NRDUCELL: NrDuCellId=0, CellRadius=10000;
//Restoring the slot configuration and slot structure (for example, when the number of GP symbols is 4)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, CellId=0, FrequencyBand=N41, SubcarrierSpacing=30KHZ, SlotAssignment=8_2_DDDDDDDSUU, SlotStructure=SS54;
//(Optional) Turning off the switch for adaptive adjustment when strong interference exists due to differences in the number of GP symbols in the slot structures between common cells and cells with Extended Cell Range enabled
MOD GNODEBALGO: ChnCalibPolSw=ADAPTIVE_ADJ_SW-0;
//(Optional) Turning off the related switch for time-domain interference avoidance when strong interference exists due to differences in the number of GP symbols in the slot structures between common cells and cells with Extended Cell Range enabled (for example, when the slot structures of common cells and cells with Extended Cell Range enabled are SS54 and SS518, respectively)
MOD GNODEBALGO: TimeIntrfAvoidSw=SS518_4GP_TIME_INTRF_AVOID_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

To verify whether Extended Cell Range of 90 km (TDD) has taken effect, perform the following operations:

1. Run the **LST NRDUCELL** command to check whether the **NRDUCell**.***CellRadius*** parameter value is greater than **60000**.
2. Run the **DSP NRDUCELL** command to check the cell status and verify that the cell has been activated.

5.4.3 Network Monitoring

Monitor the network running status using performance indicators. For details, see **5.2.2 Impacts**.

6 Directional Outmigration of Cell-Center UEs from an Extended-Range Cell

6.1 Principles

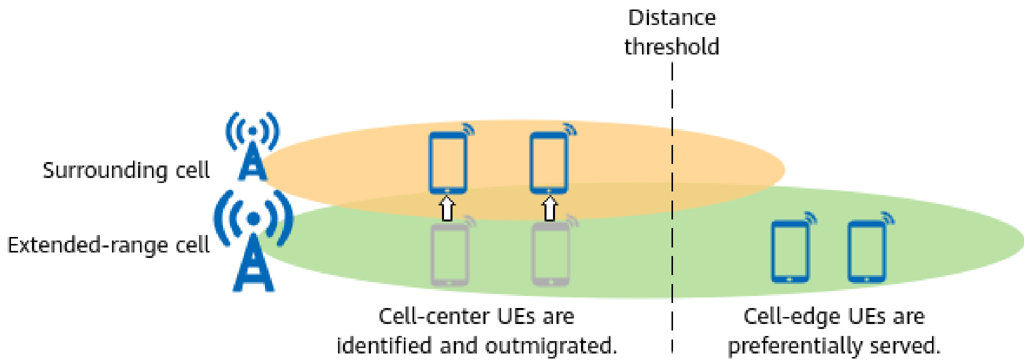
Extended-range cells refer to cells with a radius greater than 3 km.

Generally, extended-range cells require high-altitude sites and antennas to provide long-distance coverage. This makes selecting suitable sites for these cells challenging. In addition, extended-range cells often overlap with surrounding cells. Against this backdrop, the function of directional outmigration of cell-center UEs from an extended-range cell is introduced, particularly for sea coverage and emergency sites, to preferentially ensure a good user experience for cell-center UEs in extended-range cells. It is controlled by the **VIRTUAL_DEDICATED_NET_SW** option of the **NRCellAlgoSwitch**.***ExtendedCellRangePolSw*** parameter. With this function, as shown in **Figure 6-1**, the gNodeB identifies cell-center UEs in an extended-range cell and migrates them directionally to surrounding cells, allowing the extended-range cell to preferentially serve its cell-edge UEs. The gNodeB only outmigrates target UEs when all of the following requirements are met:

- UEs must be identified as cell-center UEs. Specifically, UEs are considered as cell-center UEs when the distance between them and the gNodeB is less than a certain threshold specified by the **NRCellMobilityConfig**.***CellCenterUserDisThld*** parameter.
- If multiple SSB frequencies are configured for extended-range cells that provide contiguous coverage, the **NRCellFreqRelation**.***ExtendedRangeCellFlag*** parameter must be set to **EXTENDED_RANGE** for SSB frequencies that differ from the SSB frequency of the local cell. The gNodeB will filter out frequencies with the **EXTENDED_RANGE** setting when delivering inter-frequency measurement configurations to cell-center UEs in the local cell. This prevents UEs from migrating between extended-range cells.
- Cell-center UEs must send event A5 measurement reports. Specifically, cell-center UEs send event A5 measurement reports when the measured RSRP value of a neighboring cell is greater than a certain threshold throughout a specified period of time. The threshold indicates RSRP threshold 2 for event

A5 related to inter-frequency handovers for directional outmigration of cell-center UEs from an extended-range cell and is specified by the **NRCellInterFHoMeaGrp**.***CenterUeInterFA5RsrpThld2*** parameter.

Figure 6-1 Principle of directional outmigration of cell-center UEs from an extended-range cell



Directional outmigration of cell-center UEs from an extended-range cell involves coverage-based measurement-based inter-frequency handover. (For details about this handover, see descriptions of measurement-based inter-frequency handover in *SA Mobility Management in Connected Mode*.) Therefore, ensure that the following conditions are met before enabling this function:

- Target UEs in the extended-range cell (serving cell) receive signals from both the serving cell and one of its surrounding cells, which is the target cell for UE transfer. This means that there is overlapping coverage between these cells.
- The SSB frequency of the extended-range cell (serving cell) differs from the SSB frequency of one of its surrounding cells, which is the target cell for UE transfer.

This function is designed for scenarios requiring guaranteed user experience for cell-center UEs, such as sea coverage and emergency sites.

6.2 Network Analysis

6.2.1 Benefits

With this function in effect, cell-center UEs in an extended-range cell will migrate to its surrounding cells. This prevents these UEs from occupying air interface resources of the cell and thus allows these resources to be preferentially allocated to cell-edge UEs.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

This function has the following network impacts:

- For an extended-range cell:

  The proportion of cell-center UEs decreases, while the proportion of cell-edge UEs increases. The number of UEs in the cell fluctuates.
- For surrounding common cells:
  - As UEs directionally migrate from an extended-range cell to its surrounding common cells, the number of UEs and service load increase in these common cells. In addition, **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** slightly decrease or remain unchanged in light-load scenarios and decrease in heavy-load scenarios.
  - The number of handovers increases, and the non-contention-based access success rate fluctuates.

6.3 Requirements

6.3.1 Licenses

None

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
TDD	Cell Radius	**NRDUCell**.***CellRadius***	None	Directional outmigration of cell-center UEs from an extended-range cell can be enabled only when the **NRDUCell**.***CellRadius*** parameter is set to a value greater than **3000**.

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Directional outmigration of cell-center UEs from an extended-range cell takes effect for UEs in their PCells, but not in their SCells.
TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
TDD	Downlink inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Networking

- An extended-range cell (serving cell) must overlap with one of its surrounding cells, which is the target cell for UE transfer. That is, they share neighbor relationships.
- The SSB frequency of the extended-range cell (serving cell) must differ from the SSB frequency of one of its surrounding cells, which is the target cell for UE transfer.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Extended Cell Range Policy Switch	**NRCellAlgoSwitch**.***ExtendedCellRangePolSw***	Select the **VIRTUAL_DEDICATED_NET_SW** option if directional outmigration of cell-center UEs from an extended-range cell is required.
SSB Frequency Position	**NRCellFreqRelation**.***SsbFreqPos***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Frequency Band	**NRCellFreqRelation**.***FrequencyBand***	Set this parameter based on the network plan.
Extended Range Cell Flag	**NRCellFreqRelation**.***ExtendedRangeCellFlag***	• If multiple SSB frequencies are configured for extended-range cells that provide continuous coverage, this parameter must be set to **EXTENDED_RANGE** for SSB frequencies that differ from the SSB frequency of the local cell. • If one SSB frequency is shared by extended-range cells that provide continuous coverage, this parameter does not need to be set.
Cell-Center User Distance Threshold	**NRCellMobilityConfig**.***CellCenterUserDisThld***	Set this parameter based on the network plan.
Cell-Center UE Out Inter-Freq A5 RSRP Thld2	**NRCellInterFHoMeaGrp**.***CenterUeInterFA5RsrpThld2***	For an extended-range cell, it is recommended that this parameter be set to a value 1–3 dB greater than the sum of the values of the **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** and **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameters set for its surrounding common cells.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the distance threshold for determining cell-center UEs (using the distance of 5 km as an example)
MOD NRCELLMOBILITYCONFIG: NrCellId=0, CellCenterUserDisThld=5000;
//Enabling directional outmigration of cell-center UEs from an extended-range cell
MOD NRCELLALGOSWITCH: NrCellId=0, ExtendedCellRangePolSw=VIRTUAL_DEDICATED_NET_SW-1;
//(Optional) Configuring RSRP threshold 2 for event A5 related to handovers for directional outmigration of cell-center UEs. For an extended-range cell, it is recommended that this threshold be reconfigured to be 1–3 dB higher than the sum of the values of NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld and NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs set for its surrounding common cells, unless the current threshold already slightly exceeds this sum.
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1, CenterUeInterFA5RsrpThld2=-108;
//(Optional) Configuring the extended-range cell flag for the SSB frequency of the extended-range cell if this frequency is different from that of another extended-range cell considered as the local cell and these cells provide contiguous coverage (taking the band being n77 and the SSB frequency-domain position being 7719 as an example)
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=7719, FrequencyBand=N77, ExtendedRangeCellFlag=EXTENDED_RANGE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling directional outmigration of cell-center UEs from an extended-range cell
MOD NRCELLALGOSWITCH: NrCellId=0, ExtendedCellRangePolSw=VIRTUAL_DEDICATED_NET_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

After this function is enabled, observe the distribution of the numbers of random access times when the TA values are in specific ranges, which are measured using counters in **Table 6-2**. If the proportion of the numbers of random access times when the TA values are in high ranges increases, this function has taken effect.

Table 6-2 Numbers of random access times when the TA values are in specific ranges in a cell

Counter ID	Counter Name
1911817097 to 1911817109	**N.RA.TA.UE.Index0** to **N.RA.TA.UE.Index12**

6.4.3 Network Monitoring

Monitor the network running status using performance indicators. For details, see **6.2.2 Impacts**.

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- *Hyper Cell*
- *Standards Compliance*
- *High Speed Mobility*
- *Energy Conservation and Emission Reduction*
- *Remote Interference Management (Low-Frequency TDD)*
- *Cell Combination*
- *Synchronization*
- *UE Power Saving*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Co-MM (Low-Frequency TDD)*
- *Positioning*
- *Fusion Cell (Low-Frequency TDD)*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *License Control Item Lists*
- *RedCap*
- *Beam Management*
- *SA Mobility Management in Connected Mode*

